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Guideline for the clinical application of transcranial alternating current stimulation

Penghui Song[1], Xiaotong Yang[2,3], Nuo Yang[1], Yue Dong[2,3], Yuping Wang[1,2,3]; Chinese Medical Association Neurology Branch of Neuromodulation Group; Chinese Medical Association Neurology Branch of Sleep Disorders Group; Chinese Medical Association Psychosomatic Medicine Branch of Neuromodulation Group

[1]Department of Neurology, Xuanwu Hospital, Capital Medical University, Beijing 100053, China;
[2]Department of Neurology, The First Hospital of Hebei Medical University, Shijiazhuang, Hebei 050030, China;
[3]Department of Neurology, Hebei Hospital of Xuanwu Hospital, Capital Medical University, Shijiazhuang, Hebei 050030, China.

tACS的频率是另一个关键参数。低频（θ波，4-7Hz）通常用于促进放松或改善睡眠，而高频（γ波，30-100Hz）可能会增强认知功能或警觉性。

Transcranial alternating current stimulation (tACS) is a noninvasive neuromodulation technique that regulates brain function by applying sinusoidal alternating currents of specific frequencies to targeted brain regions through the scalp electrodes.[1] This guideline aims to evaluate and summarize the efficacy and safety of tACS in clinical applications. We conducted a structured literature search across major databases using the terms "tACS" or "transcranial alternating current stimulation" to identify all relevant literature up to June 2022. After rigorous screening, a total of 35 articles were included in this guideline (the detailed search and screening process is illustrated in Supplementary Figure 1, http://links.lww.com/CM9/C402). The guideline uses the Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) evidence quality evaluation and recommendation strength grading standards established by the World Health Organization (WHO) for these 35 clinical trials.[2] Following the Delphi method, electronic questionnaires were distributed to multiple specialized groups. After collection, summarization, and feedback, the application opinions and recommendation grades of tACS in various diseases were finalized (detailed procedures are provided in Supplementary Material 1, http://links.lww.com/CM9/C402).

tACS procedures: The commonly used tACS methods can be categorized into three types based on current intensity: superhigh-energy stimulation (superhigh tACS, current intensity >100 mA), high-energy stimulation (high tACS, current intensity >10–100 mA), medium-energy stimulation (medium tACS, current intensity 4–10 mA) and low-energy stimulation (low tACS, current intensity <4 mA).[3] Within a certain range, higher tACS stimulation intensities result in a broader range of endogenous effects, stronger entrainment capabilities, and deeper stimulation penetration.[4] In the current article, the same phase electrodes are included in "[]", whereas different phase electrodes/groups of electrodes are separated by "-". If multiple current loops exist, they are separated by ";". The frequency of tACS is another critical parameter. Low frequencies (θ waves, 4–7 Hz) are often used to promote relaxation or improve sleep, whereas higher frequencies (γ waves, 30–100 Hz) may enhance cognitive functions or alertness. Detailed protocols and operational procedures for tACS are described in Supplementary Material 2, http://links.lww.com/CM9/C402.

Clinical applications of tACS: ***Insomnia disorder***: tACS can modulate cortical excitability and brain function, improve sleep efficiency and quality, and alleviate insomnia symptoms by influencing the synchronization and desynchronization of neuronal electrical activity in the brain. High-quality evidence indicates that 15 mA, 77.5 Hz tACS with electrodes placed at [prefrontal lobe]-[bilateral mastoid] improves sleep quality in chronic insomnia patients. Low-quality evidence suggests that 0.75 mA, 0.75 Hz tACS with electrodes placed at [F3]-[left mastoid area]; [F4]-[right mastoid area] may also improve sleep quality in chronic insomnia patients (detailed information is provided in Supplementary Table 1, http://links.lww.com/CM9/C402).

Recommendations: For chronic insomnia patients, the application of high tACS with electrodes placed on [prefrontal region]-[bilateral mastoid] is strongly recommended to improve sleep efficiency and quality. Low tACS with electrodes on [F3]-[left mastoid area]; [F4]-[right mastoid area] might also be useful in improving sleep quality.

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Correspondence to: Yuping Wang, Department of Neurology, Xuanwu Hospital, Capital Medical University, Beijing 100053, China; Department of Neurology, The First Hospital of Hebei Medical University; Hebei Hospital of Xuanwu Hospital, Capital Medical University, Shijiazhuang, Hebei 050030, China
E-Mail: wangyuping01@sina.cn

王玉平，首都医科大学宣武医院神经内科

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常用的tACS方法根据电流强度可分为三种类型：超高能刺激（超高 tACS，电流强度 >100 mA）、高能刺激（高 tACS，电流强度 >10 – 100 mA）、中等能量刺激（中等 tACS，电流强度4-10mA）和低能量刺激（低 tACS，电流强度<4mA）。

tACS的临床应用：失眠障碍：tACS可以通过影响大脑中神经元电活动的同步和不同步来调节皮层兴奋性和脑功能，提高睡眠效率和质量，缓解失眠症状。高质量的证据表明，将电极放置在[前额叶]-[双侧乳突]的15mA，77.5Hz的tACS可改善慢性失眠患者的睡眠质量。低质量证据表明，0.75mA、0.75Hz的tACS，电极放置在[F3]-[左乳突区域];[F4]-[右乳突区]也可能改善慢性失眠患者的睡眠质量。

认知功能障碍和阿尔茨海默病

Cognitive dysfunction and Alzheimer's disease (AD): Clinical studies on tACS for cognitive dysfunction have primarily focused on patients with mild cognitive impairment (MCI) or mild-to-moderate dementia. Some studies included patients who were diagnosed definitively with AD through pathologic examination. High-quality evidence indicates that 2 mA, 40 Hz tACS placed over the bilateral temporal regions enhances cognitive functions in mild-to-moderate AD patients. Moderate-quality evidence suggests that 3 mA, 40 Hz tACS placed over [Pz]-[right deltoid] improves episodic memory function in MCI and AD patients. Some low-quality evidence did not find a therapeutic effect detectable by cognitive scales but may reduce the p-Tau burden. tACS combined with cognitive training may improve memory function for a longer duration than cognitive training alone. tACS combined with 40 Hz sound stimulation for 3 weeks may improve cognitive function in AD patients. Fourteen weeks of home-based tACS treatment may improve cognitive function in AD-related dementia patients. In addition, tACS appears to offer superior cognitive benefits compared with transcranial direct current stimulation (tDCS) in MCI patients (detailed information is provided in Supplementary Table 2, http://links.lww.com/CM9/C402).

为期14周的家庭tACS治疗可能会改善AD相关痴呆患者的认知功能。此外，与MCI患者的经颅直流电刺激（tDCS）相比，tACS似乎提供了更好的认知益处。

对于MCI或轻度至中度AD患者，强烈建议在双侧颞区应用低tACS和电极，以改善认知功能。

Recommendations: For MCI or mild-to-moderate AD patients, the application of low tACS with electrodes in the bilateral temporal region is strongly recommended to improve cognitive function. Low tACS with electrodes on [Pz]-[right deltoid] is also recommended.

Chronic pain: Moderate-quality evidence indicates that 10 Hz tACS placed on [F3, F4]-[Pz] alleviates chronic back pain. Moderate-quality evidence suggests that 1–2 mA tACS at 4 Hz or 30 Hz (depending on the individual's dominant frequency spectral power) placed on [the highest power spectrum difference area]-[ipsilateral mastoid] could reduce pain in fibromyalgia syndrome patients. Low-level evidence indicates that acute application of 0.4 mA, 140 Hz tACS placed on [Oz]-[Cz] for 15 min may effectively terminate migraine attacks (detailed information is provided in Supplementary Table 3, http://links.lww.com/CM9/C402).

Recommendations: For chronic back pain patients, the application of low tACS with electrodes placed on [F3, F4]-[Pz] is recommended to alleviate pain. For fibromyalgia syndrome patients, the application of low tACS with electrodes placed on [the highest power spectrum difference area]-[ipsilateral mastoid] is weakly recommended to improve symptoms. For migraine patients, the application of low tACS with electrodes placed on [Oz] – [Cz] to terminate migraine attacks is weakly recommended.

Vestibular disorder: Moderate-quality evidence indicates that individual alpha frequency tACS placed on [prefrontal]-[occipital] (2 mA) or [prefrontal, occipital]-[left upper arm] (4 mA) for 20 min per session over 10–12 sessions alleviates oscillatory vertigo in Mal de Débarquement Syndrome (MdDS) patients. The former configuration is more effective, and stimulating at a slightly higher frequency than the individual alpha frequency is preferable (detailed information is provided in Supplementary Table 4, http://links.lww.com/CM9/C402).

Recommendations: For MdDS patients, the application of low tACS with electrodes placed on [prefrontal]-[occipital] is weakly recommended to alleviate oscillatory vertigo. Stimulating at a frequency slightly higher than the individual alpha frequency is preferable.

Stroke rehabilitation: Low-quality evidence suggests that 0.4 mA 20 Hz tACS placed on the bilateral mastoid for 30 min per session over 15 sessions across 3 weeks, combined with rehabilitation training, may improve overall functional recovery in patients with subacute supratentorial intracerebral hemorrhage or infarction (detailed information is provided in Supplementary Table 5, http://links.lww.com/CM9/C402).

Recommendations: For subacute-stage stroke patients, the application of tACS with electrodes placed on the bilateral mastoid combined with rehabilitation training is weakly recommended to improve overall functional recovery.

Parkinson's disease: Moderate-quality evidence indicates that combining 1–2 mA, 4 Hz tACS placed on [the power spectral difference area]-[ipsilateral mastoid] with rehabilitation for 30 min per session over 10 sessions across 2 weeks enhances motor and cognitive functions. Furthermore, 15 mA 77.5 Hz tACS with the electrodes placed on [forehead]-[bilateral mastoid area] for 45 min per session for over 10 sessions across 2 weeks has no significant effects on mood or sleep (detailed information is provided in Supplementary Table 6, http://links.lww.com/CM9/C402).

Recommendations: For Parkinson's disease patients, the application of 4 Hz tACS with electrodes on [the power spectral difference area]-[ipsilateral mastoid] combined with rehabilitation training is recommended to improve motor and cognitive functions.

Epilepsy: Moderate-quality evidence indicates that 2 mA, 3 Hz tACS with electrodes on [the most frequent interictal epileptiform discharge area]-[ipsilateral supraorbital region] for 30 or 60 min over 3 or 5 days fails to reduce seizure frequency in patients with multifocal refractory epilepsy (including temporal lobe epilepsy, frontal lobe epilepsy, and Lennox–Gastaut syndrome). Very low-quality evidence suggests that 1 mA, 3 Hz tACS placed on [Fp1]-[Fp2] for 60 min daily over 4 days increased the seizure frequency by 75% in a 16-year-old female with myoclonic epilepsy, followed by a 15-day seizure-free period (detailed information is provided in Supplementary Table 7, http://links.lww.com/CM9/C402).

Recommendations: No specific recommendations are provided for tACS in epilepsy due to insufficient evidence.

Depressive disorders (including depressive states): High-quality evidence from three studies indicates that 15 mA, 77.5 Hz tACS with electrodes on the [prefrontal lobe]-[bilateral mastoid] for 40 min daily over 4 weeks significantly alleviates depressive symptoms. Moderate-quality evidence shows that 1 mA, 10 Hz tACS with electrodes on [F3, F4]-[Cz] for 40 min daily over 5 days yields better antidepressant outcomes than 40 Hz tACS. Very low-quality evidence suggests that 2 mA, 40 Hz tACS with electrodes on [F3]-[F4] for 10 min twice daily

三项研究的高质量证据表明，在4周内，每天在[前额叶]-[双侧乳突]上放置15mA、77.5Hz的tACS电极40分钟，可显著缓解抑郁症状。

over 10 days significantly reduces depressive symptoms (detailed information is provided in Supplementary Table 8, http://links.lww.com/CM9/C402).

Recommendations: For patients with depression, the application of high tACS with electrodes on the [prefrontal lobe]-[bilateral mastoid] is strongly recommended to alleviate depressive symptoms. Low tACS with electrodes on [F3, F4]-[Cz] is also recommended. Combining tACS with antidepressants is advised to enhance therapeutic outcomes.

Schizophrenia (including psychotic symptoms): Moderate-quality evidence indicates that 2 mA, 10 Hz tACS with electrodes on [the region between F3 and Fp1, the region between T3 and P3]-[Cz] for 20 min per session over 10 sessions may improve auditory hallucinations. Another moderate quality of evidence suggests that 2 mA, 6 Hz tACS with electrodes on [F1, F5, AF3, FC3]-[CPz]; [P1, P5, CP3, PO3]-[FCz] alleviates negative symptoms in schizophrenia patients. Furthermore, 2 mA, 10 Hz tACS on [AFz]-[Cz] may reduce the severity of delusions, and 2 mA, 6 Hz and 40 Hz tACS on [F3]-[P3] may improve working memory in schizophrenia patients (detailed information is provided in Supplementary Table 9, http://links.lww.com/CM9/C402).

Recommendations: For schizophrenia patients, tACS with electrodes on [the region between F3 and Fp1, the region between T3 and P3]-[Cz] is weakly recommended to improve auditory hallucinations. tACS with electrodes on [F1, F5, AF3, FC3]-[CPz]; [P1, P5, CP3, PO3]-[FCz] is recommended to alleviate negative symptoms.

Substance addiction: Low-quality evidence from three studies suggests that low tACS with electrodes placed on [F3, F4]-[Cz] (2 mA, 10 Hz) for 30 or 40 min may reduce habitual behaviors in individuals with substance addiction (detailed information is provided in Supplementary Table 10, http://links.lww.com/CM9/C402).

Recommendations: No specific recommendations are provided for tACS in substance addiction due to insufficient evidence.

Tolerability and safety of tACS therapy: The overall tolerability and safety of tACS are favorable. Common mild adverse reactions including photophobia (flash, phosphenes), itching, burning, tingling, fever, dizziness, and headache, with tingling, itching, and burning being the most frequently reported. Supplementary Material 3 and Supplementary Table 11, http://links.lww.com/CM9/C402 summarize the common adverse effects observed during tACS therapy and provide recommended management strategies. Supplementary Table 12, http://links.lww.com/CM9/C402 outlines recommendations for tACS application in specific populations. Furthermore, continuous monitoring of participants' responses during tACS administration is strongly advised, and meticulous inspection of the electrode-skin interface should be performed before and after each session.

Prospects for tACS: With advancements in neuroscience, future research is expected to establish disease-specific frequency-intensity-target models using multimodal technologies and to develop personalized treatment strategies based on individual neural oscillations and genetic characteristics. The widespread clinical application of tACS will ultimately depend on rigorous evidence-based validation and the establishment of standardized technical guidelines.

Members of the Expert Steering Group (sorted by last name)

Dongmei An, Xuebing Cao, Haobo Chen, Ling Chen, Guiyun Cui, Jiahui Deng, Wei Deng, Jing Ding, Tao Feng, Zhengzhi Feng, Yuan Geng, Qiang Guan, Yi Guo, Yue Hou, Shaohua Hu, Yonghua Huang, Li Kuang, Tongliang Li, Zhe Li, Zhanhua Liang, Jun Liu, Xuedong Liu, Yiming Liu, Xiaoyan Long, Jinghong Ma, Yan Mei, Xianghong Meng, Yijun Song, Chaoxia Su, Shujuan Tian, Yuling Tian, Zhirong Wan, Chunxue Wang, Jijun Wang, Kai Wang, Lin Wang, Mengyang Wang, Mingwei Wang, Qing Wang, Qun Wang, Xiaorong Wang, Yanyong Wang, Yuping Wang, Zan Wang, Shenhong Weng, Bingjie Wu, Jianjun Wu, Wenqing Wu, Yuncheng Wu, Anmu Xie, Fengquan Xu, Yuehong Xu, Zhi Xu, Liqing Yan, Xinling Yang, Qinyong Ye, Jie Yuan, Yonggui Yuan, Lingyun Zeng, Linwei Zhang, Xiong Zhang, Yan Zhang, Yang Zhang, Yuhu Zhang, Peng Zheng, Wei Zheng, Di Zhong, Bo Zhou, Hongcan Zhu.

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Conflicts of interest

None.

References

1. Wischnewski M, Alekseichuk I, Opitz A. Neurocognitive, physiological, and biophysical effects of transcranial alternating current stimulation. Trends Cogn Sci 2023;27:189–205. doi: 10.1016/j.tics.2022.11.013.
2. Langer G, Meerpohl JJ, Perleth M, Gartlehner G, Kaminski-Hartenthaler A, Schünemann H. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. Z Evid Fortbild Qual Gesundhwes 2012;106:357–368. doi: 10.1016/j.zefq.2012.05.017.
3. Antal A, Alekseichuk I, Bikson M, Brockmöller J, Brunoni AR, Chen R, *et al*. Low intensity transcranial electric stimulation: Safety, ethical, legal regulatory and application guidelines. Clin Neurophysiol 2017;128:1774–1809. doi: 10.1016/j.clinph.2017.06.001.
4. Kehler L, Francisco CO, Uehara MA, Moussavi Z. The effect of transcranial alternating current stimulation (tACS) on cognitive function in older adults with dementia. Annu Int Conf IEEE Eng Med Biol Soc 2020;2020:3649–3653. doi: 10.1109/EMBC44109.2020.9175903.

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tACS治疗的耐受性和安全性：tACS的总体耐受性和安全性是有利的。常见的轻度不良反应包括畏光（闪光、磷化）、瘙痒、灼热、刺痛、发热、头晕和头痛，其中刺痛、瘙痒和灼热是最常见的报告。